D06 · Logistics Governance & Analytics

03 · Logistics risk management & supply chain resilience

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Logistics risk management identifies the disruptions that could break the physical network — carrier capacity loss, border or port closures, facility incidents, cyber events, extreme weather — and prepares tested alternatives and response plans for them.

2 WHY IT MATTERS

Resilience is not built during a crisis; it is built before. **The speed of response depends on alternatives that are already qualified and decision rights that are already clear (Sheffi — The Resilient Enterprise, MIT Press)**. A business continuity plan that has never been exercised is a document, not a capability.

3 HOW IT IS DONE

1. **Map risks by probability and impact.** Include the uncomfortable ones: primary 3PL failure, cyber events, border closures.
2. **Pre-qualify alternative capacity.** Backup carriers, facilities and modes, contracted or pre-negotiated.
3. **Write activation triggers and roles.** Who decides, when, and with which pre-approved budget.
4. **Exercise the plan every year.** Tabletop or live simulations that expose gaps before a real event does.

IN PRACTICE

Map the risks that feel improbable — they are often the most expensive

Primary 3PL failure, major border closure and cyber events that freeze systems belong on the map even if they feel remote.

Run at least one simulation or tabletop exercise per year

A BCP that has never been practiced is a document, not an operational capability.

Extend the BCP logic to critical 3PLs and carriers

Your resilience is limited by the resilience of the partners that execute the physical network.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Peak-season capacity failure of the primary carrier (45% of outbound volume)

BCP Element	Activation Time	Outcome
Backup carrier A (pre-qualified)	2 h	Absorbed 20% of volume at +18% rate
Backup carrier B	4 h	Absorbed 15% at +28% rate
Customer prioritization (ABC)	3 h	Strategic customers protected
Negotiation with primary for partial recovery	6 h	Additional 10% recovered

Result: Peak-week OTIF held at 88% versus an estimated 45–50% without the BCP. Premium cost was real but far smaller than the commercial and reputational cost of a full service collapse.

HOW IT IS MEASURED

BCP Activation Time

Hours from confirmed disruption to activation of documented alternatives

Benchmark: <4 h for high-probability scenarios · <8 h for less frequent critical scenarios

BCP Coverage %

(Critical risks with documented and tested alternatives / Total critical risks) × 100

Benchmark: >90% of material risks should have a prepared response

WHAT YOU WOULD USE

Risk & Resilience

Everstream Analytics / Resilinc / similar risk-intelligence platforms — Provide early warning on disruptions affecting suppliers, lanes and nodes.

Control tower + scenario tools (Kinaxis, Blue Yonder, etc.) — Enable rapid impact assessment and response coordination when events occur.

4 SO WHAT

Decide the response before the event. Map the risks, qualify alternatives in advance, make activation rules explicit and rehearse the plan at least once a year.

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